Data Wrangling & Tidy Data

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~80%

of time is spent cleaning and preparing data before analysis.

Dasu T, Johnson T (2003). Exploratory Data Mining and Data Cleaning. John Wiley & Sons.

Agenda

Definitions

Tidy Data, Data Wrangling, Tools

Steps of data wrangling

Reshaping, Combine, Handling Missing Data, Transformation and Normalization

Practice tutorial

Recognize patterns and apply the concepts

Tidy Data

is a standard way of structuring datasets to
facilitate manipulation, visualization and modeling

[1]

each **variable** is a **column**;
each **column** is a **variable**

each **observation** is a **row**;
each **row** is an **observation**.

each **value** is a **cell**;
each **cell** is a single **value**

Messy vs Tidy

Column headers are values, not variable names

[2]

[2]

religion	<$10k	$10–20k	$20–30k	$30–40k	$40–50k	$50–75k
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Don't know/refused	15	14	15	11	10	35
Evangelical Prot	575	869	1064	982	881	1486
Hindu	1	9	7	9	11	34
Historically Black Prot	228	244	236	238	197	223
Jehovah's Witness	20	27	24	24	21	30
Jewish	19	19	25	25	30	95

Messy

religion	income	freq
Agnostic	<$10k	27
Agnostic	$10–20k	34
Agnostic	$20–30k	60
Agnostic	$30–40k	81
Agnostic	$40–50k	76
Agnostic	$50–75k	137
Agnostic	$75–100k	122
Agnostic	$100–150k	109
Agnostic	>150k	84
Agnostic	Don't know/refused	96

Tidy

Messy vs Tidy

Multiple variables stored in one column

[2]

country	year	column	cases
AD	2000	m014	0
AD	2000	m1524	0
AD	2000	m2534	1
AD	2000	m3544	0
AD	2000	m4554	0
AD	2000	m5564	0
AD	2000	m65	0
AE	2000	m014	2
AE	2000	m1524	4
AE	2000	m2534	4
AE	2000	m3544	6
AE	2000	m4554	5
AE	2000	m5564	12
AE	2000	m65	10
AE	2000	f014	3

Messy

[2]

country	year	sex	age	cases
AD	2000	m	0–14	0
AD	2000	m	15–24	0
AD	2000	m	25–34	1
AD	2000	m	35–44	0
AD	2000	m	45–54	0
AD	2000	m	55–64	0
AD	2000	m	65+	0
AE	2000	m	0–14	2
AE	2000	m	15–24	4
AE	2000	m	25–34	4
AE	2000	m	35–44	6
AE	2000	m	45–54	5
AE	2000	m	55–64	12
AE	2000	m	65+	10
AE	2000	f	0-14	3

Tidy

Data Wrangling

transforms and maps data through a series of steps
to become clean, consistent, reliable and useful for its intended application.
~ibm

> **Goal: prepare data for analysis by converting it into a tidy format.**

Tools

for data wrangling

pandas

melt(), pivot(), merge(), groupby(), agg()

tidyverse

dplyr, tidyr

Data Wrangler

AI-Assisted data wrangling extension

Import & Export Data

in pandas

Import

```
pd.read_csv(),
pd.read_excel(),
pd.read_json()
```

Inspect

```
df.head(),
df.info(),
df.describe()
```

Export

```
df.to_csv(),
df.to_excel(),
df.to_json()
```

Reshape

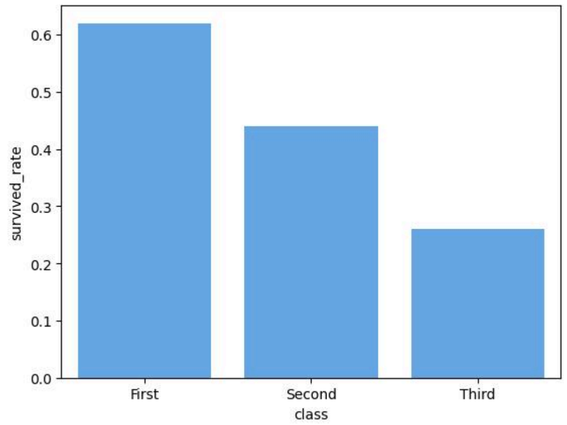
```
sns.barplot(data=df,x="class", y="survival_rate")
```

class	First	Second	Third
survival_rate	0.62	0.44	0.26

→ **Error**

class	survival_rate
First	0.62
Second	0.44
Third	0.26

→



Reshape Data

melt

[3]

```
df3
```

	first	last	height	weight
0	John	Doe	5.5	130
1	Mary	Bo	6.0	150

```
df3.melt(id_vars=['first', 'last'])
```

	first	last	variable	value
0	John	Doe	height	5.5
1	Mary	Bo	height	6.0
2	John	Doe	weight	130
3	Mary	Bo	weight	150

- **wide to long**
- **column headers into row values**

Reshape Data

pivot

edited [3]

```
df.pivot(index='foo',
         columns='bar',
         values='baz')
```

df

	foo	bar	baz
0	one	A	1
1	one	B	2
2	one	C	3
3	two	A	4
4	two	B	5
5	two	C	6

Stacked

bar	A	B	C
one	1	2	3
two	4	5	6

Record

- **long to wide**
- **sort out data where a single observation is scattered over multiple rows**

Reshape Data

stack

edited [3]

`df2`

first	second	A	B
bar	one	1	2
bar	two	3	4
baz	one	5	6
baz	two	7	8

MultiIndex

`stacked = df2.stack()`

first	second		
bar	one	A	1
bar	one	B	2
bar	two	A	3
bar	two	B	4
baz	one	A	5
baz	one	B	6
baz	two	A	7
baz	two	B	8

MultiIndex

> **similar to melt but with multi level indices**

Reshape Data

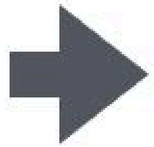
unstack

edited [3]

`stacked`

first	second		
bar	one	A	1
bar	one	B	2
bar	two	A	3
bar	two	B	4
baz	one	A	5
baz	one	B	6
baz	two	A	7
baz	two	B	8

MultiIndex



`stacked.unstack()`

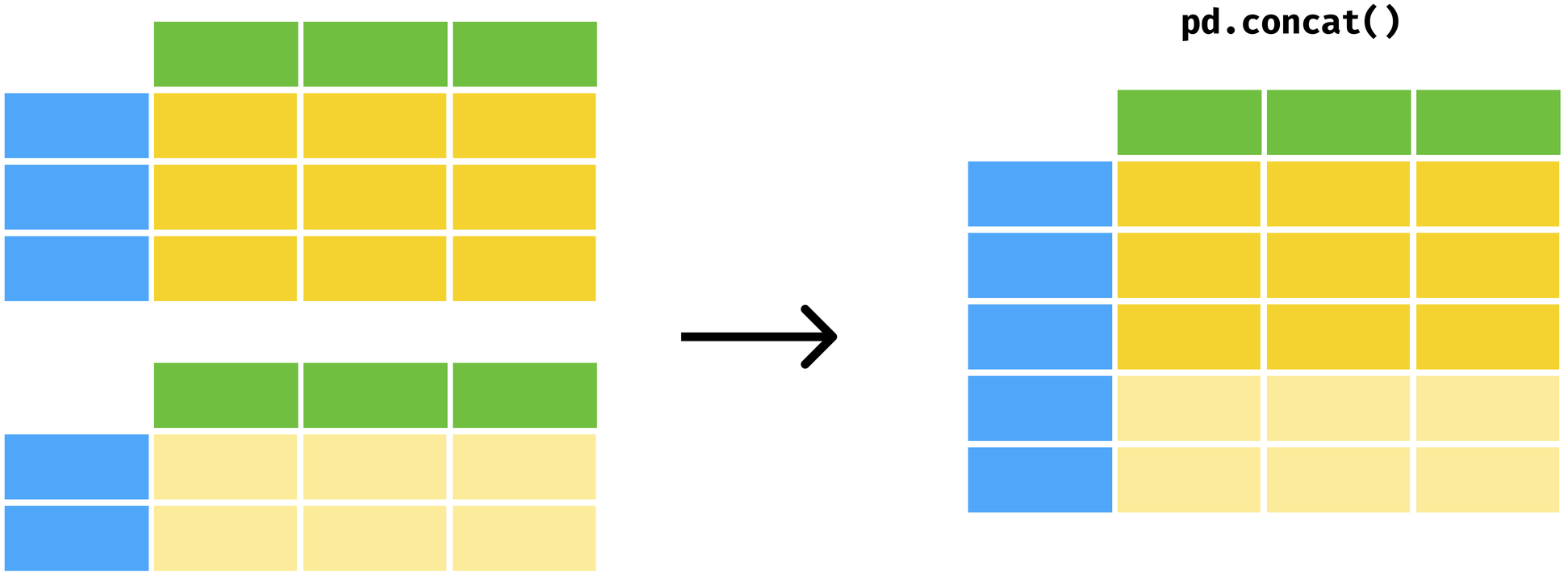
first	second	A	B
bar	one	1	2
bar	two	3	4
baz	one	5	6
baz	two	7	8

MultiIndex

> **similar to pivot but with multi level indices**

Combine Data

concat

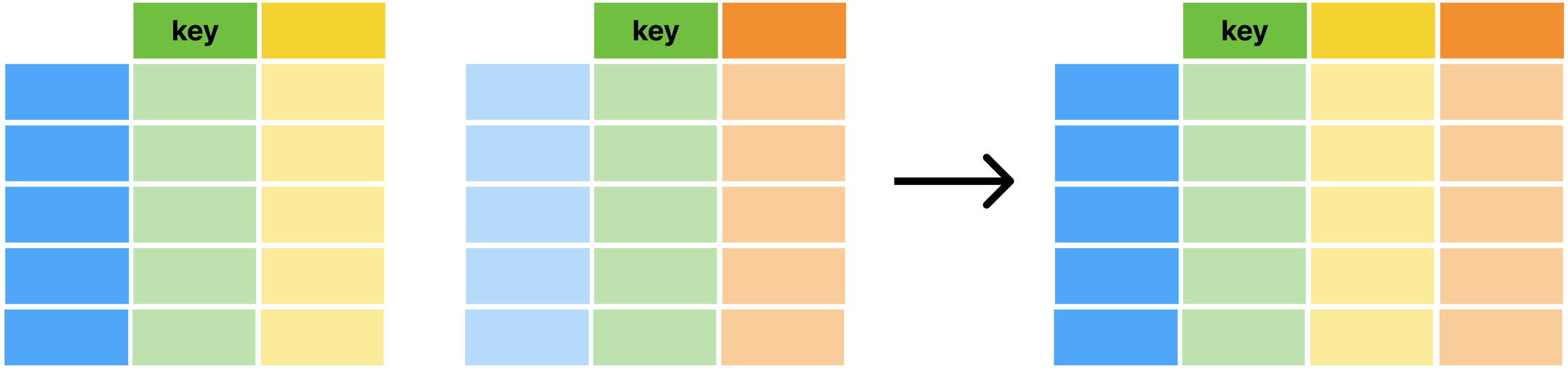


- **append similar rows (observations)**
- **two or more data frames with the same index or the same columns**

Combine Data

merge

```
pd.merge(left, right,
on=["key"], how="inner")
```



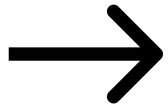
- join two datasets based on a key column
- when different tables have related data
- left / right / inner / outer

Combine / Missing data

Merging:

	customer_id	name
0	1	Alice
1	2	Bob
2	3	Carla

	customer_id	order_total
0	1	120
1	3	55
2	4	90



Results in:

	customer_id	name	order_total
0	1	Alice	120.0
1	2	Bob	NaN
2	3	Carla	55.0
3	4	NaN	90.0

Handling Missing Data

Kind of Absence

Structural

Data that cannot logically exist
(e.g. “number of pregnant males”)

Observational

Data that should exist but
wasn’t recorded

How it Appears

Explicit

Visibly missing in the dataset
→ presence of absence

Implicit

Hidden due to dataset’s structure
(e.g. category combination not present)
→ absence of presence

Explicit Missing

Common types:

→ np.nan for numeric columns
→ pd.NA for all nullable dtypes
→ None for Object dtypes

	A	B	C	D
0	1.0	2.0	apple	10.0
1	NaN	3.0	None	20.0
2	-99.0	NaN	banana	NaN
3	4.0	5.0	mango	40.0
4	5.0	6.0	pear	50.0

Fill / Impute:

```
df.fillna(0)
df.fillna(method="ffill")
df.fillna(method="bfill")
df.fillna({"A": 0, "B": 1})
```

→

	A	B	C	D
0	1.0	2.0	apple	10.0
1	0.0	3.0	0	20.0
2	-99.0	0.0	banana	0.0
3	4.0	5.0	mango	40.0
4	5.0	6.0	pear	50.0

Remove:

```
df.dropna(axis=0)
df.dropna(axis=1)
df.dropna(how="all")
df.dropna(thresh=3)
df.interpolate()
```

→

	A	B	C	D
0	1.0	2.0	apple	10.0
3	4.0	5.0	mango	40.0
4	5.0	6.0	pear	50.0

Replace coded missing values:

```
df.replace({-99: pd.NA})
```

→

	A	B	C	D
0	1.0	2.0	apple	10.0
1	NaN	3.0	None	20.0
2	<NA>	NaN	banana	NaN
3	4.0	5.0	mango	40.0
4	5.0	6.0	pear	50.0

Implicit Missing

	year	qtr	price
0	2020	1	1.88
1	2020	2	0.59
2	2020	3	0.35
3	2020	4	NaN
4	2021	2	0.92
5	2021	3	0.17
6	2021	4	2.66

qtr	1	2	3	4
year				
2020	1.88	0.59	0.35	NaN
2021	NaN	0.92	0.17	2.66

```
df.pivot(columns="qtr",
         values="price",
         index="year")
```

Two missing observations:

Explicitly missing → price for Q4 2020
Implicitly missing → Q1 2021

Make them explicit

Use pivoting, reshaping or re-indexing to reveal gaps

Missing Values in Categorical Variables

	name	smoker	age
0	Ikaia	no	34
1	Oletta	no	88
2	Leriah	previously	75
3	Dashay	no	47
4	Tresaun	yes	56

	name	smoker	age
0	Ikaia	no	34
1	Oletta	no	88
2	Leriah	previously	75
3	Dashay	no	47

value_counts()

```
smoker
no            3
previously    1
yes           0
Name: count, dtype: int64
```

groupby()

```
smoker
no            56.333333
previously    75.000000
yes                 NaN
Name: age, dtype: float64
```

Can be used to remove categories:

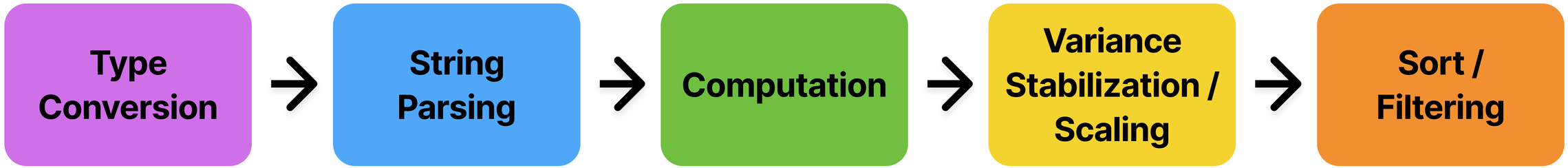
```
df_cut["smoker"].cat
.remove_unused_categories()
```

```
smoker
no            3
previously    1
Name: count, dtype: int64
```

Transformation

“Changing data values to make them suitable for analysis”

> **Modifying or creating new variables to enhance or prepare data for modeling.**



Case Study: Transforming a Weather Dataset

	id	year	month	element	d1	d2	d3
0	MX17004	2010	1	tmax	27.0	24.0	None
1	MX17004	2010	1	tmin	14.0	10.0	None
2	MX17004	2010	1	tmax	NaN	NaN	None

Messy (Wide) Data

```
tidy_weather = (
    weather
    .melt(
        id_vars=["id", "year", "month", "element"],
        var_name="day",
        value_name="value",
    )

)
tidy_weather.loc[:, "day"] = (
    tidy_weather["day"]
    .str.extract("(\d+)")
    .astype(float)
)
tidy_weather = (
    tidy_weather
    .dropna(subset=["value"])
    .sort_values(["id", "year", "month", "day"])
)
```

Transformation Steps

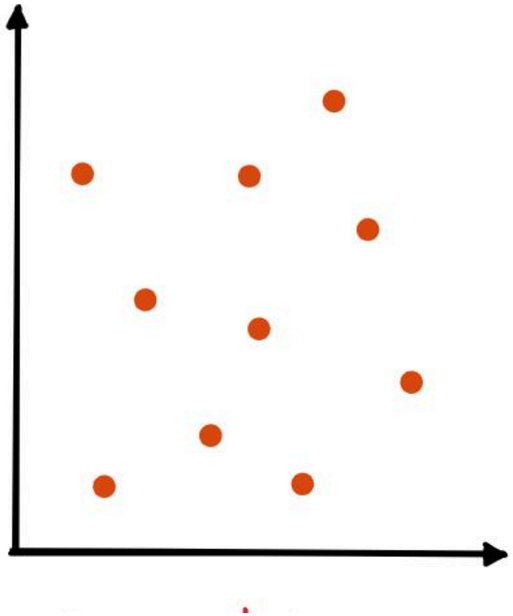
	id	year	month	element	day	value
0	MX17004	2010	1	tmax	1.0	27.0
1	MX17004	2010	1	tmin	1.0	14.0
3	MX17004	2010	1	tmax	2.0	24.0
4	MX17004	2010	1	tmin	2.0	10.0

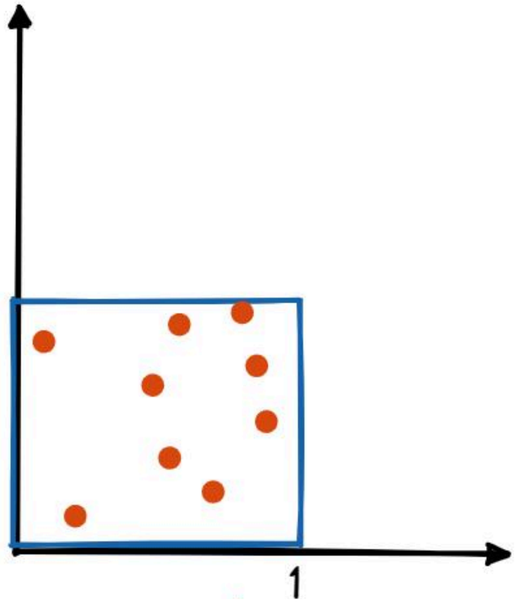
Tidy (Long) Data

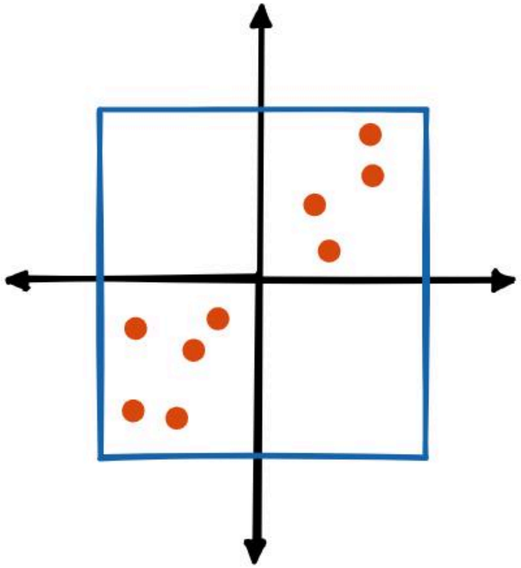
Normalization

scaling numeric values into a common range

[4]







required for a lot of tasks
e.g machine learning

Min–Max Normalization

Keeps relative differences
(0-1)

Z-Score Standardization

Centers data at 0 with unit variance
(e.g. −3 to +3)

Extra: AI Assisted Data Wrangling

Automating tedious transformations with AI

Data Wrangler Extension

Uses ML

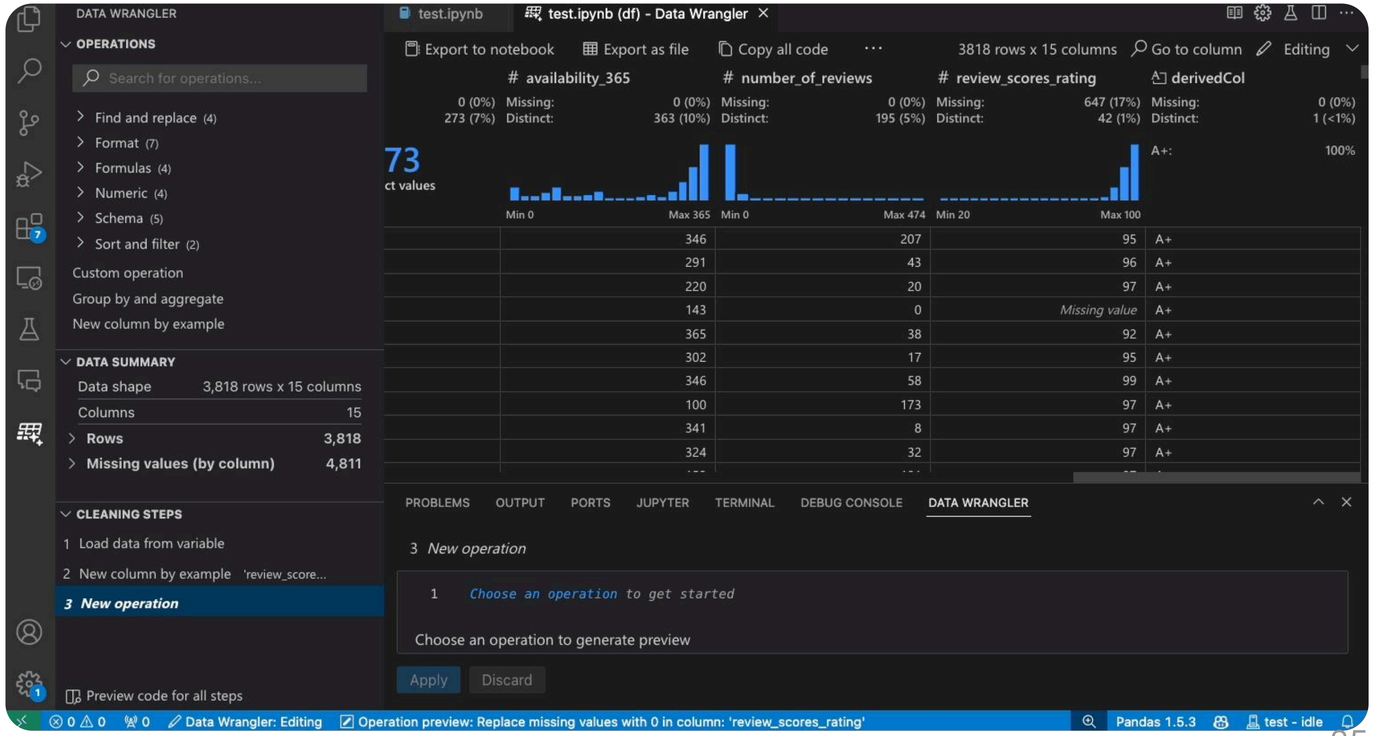
To automatically suggest Pandas transformations

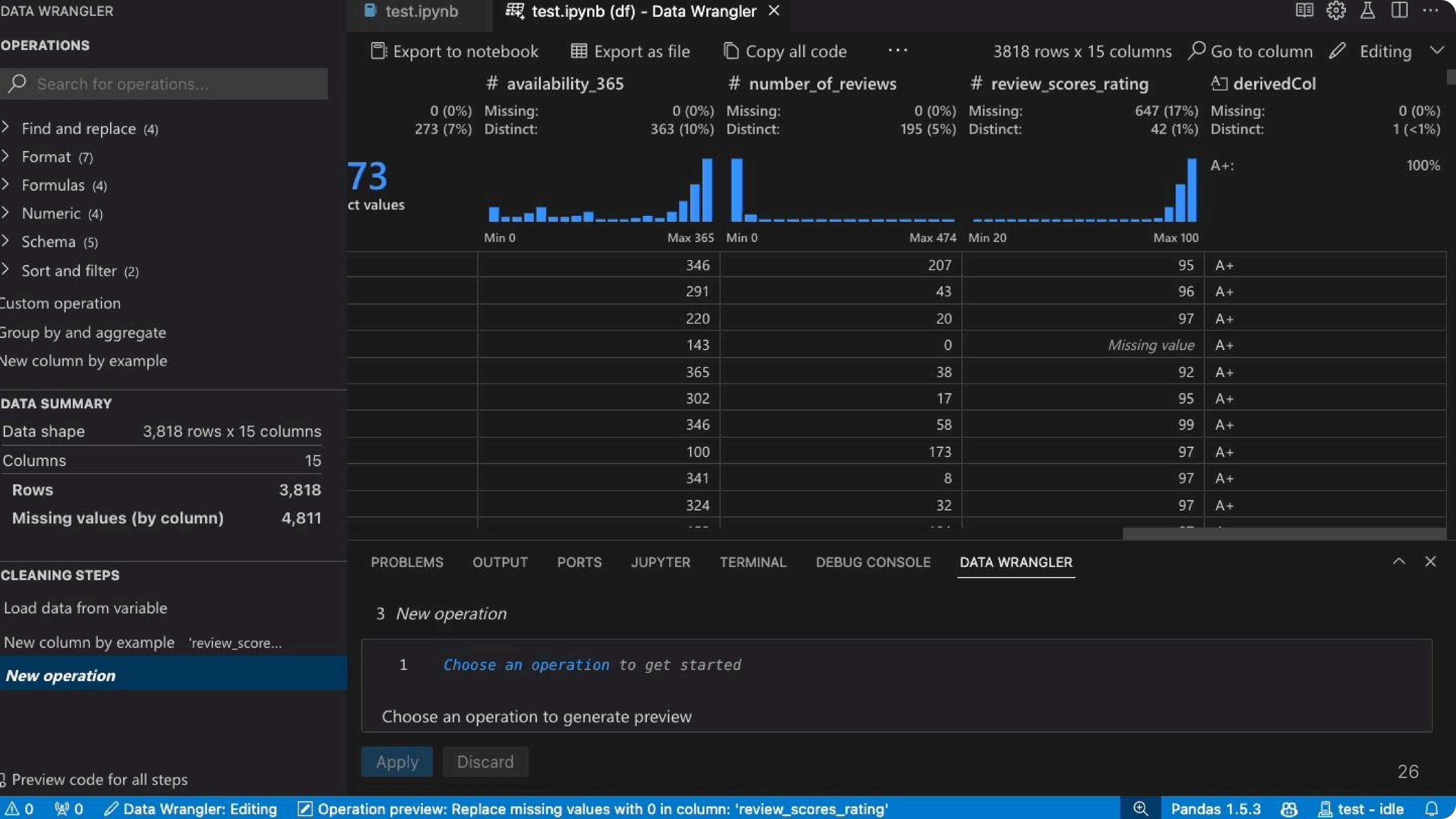
Supports multiple operations

Like type conversion, column renaming, filtering, grouping, NA handling

Generates Code

Clean Python / Pandas code for instant reproducibility




DATA WRANGLER
OPERATIONS
Search for operations...
Find and replace (4)
Format (7)
Formulas (4)
Numeric (4)
Schema (5)
Sort and filter (2)
Custom operation
Group by and aggregate
New column by example
DATA SUMMARY
Data shape 3,818 rows x 15 columns
Columns 15
Rows 3,818
Missing values (by column) 4,811
CLEANING STEPS
Load data from variable
New column by example 'review_score...
New operation
Preview code for all steps
test.ipynb
test.ipynb (df) - Data Wrangler
Export to notebook
Export as file
Copy all code
3818 rows x 15 columns
Go to column
Editing
availability_365
number_of_reviews
review_scores_rating
derivedCol
0 (0%) Missing:
273 (7%) Distinct:
0 (0%) Missing:
363 (10%) Distinct:
0 (0%) Missing:
195 (5%) Distinct:
647 (17%) Missing:
42 (1%) Distinct:
0 (0%) Missing:
1 (<1%) Distinct:
73
ct values
A+: 100%
Min 0 Max 365
Min 0 Max 474
Min 20 Max 100
346 207 95 A+
291 43 96 A+
220 20 97 A+
143 0 Missing value A+
365 38 92 A+
302 17 95 A+
346 58 99 A+
100 173 97 A+
341 8 97 A+
324 32 97 A+
PROBLEMS
OUTPUT
PORTS
JUPYTER
TERMINAL
DEBUG CONSOLE
DATA WRANGLER
3 New operation
1 Choose an operation to get started
Choose an operation to generate preview
Apply
Discard
26
0
0
Data Wrangler: Editing
Operation preview: Replace missing values with 0 in column: 'review_scores_rating'
Pandas 1.5.3
test - idle

Summary

Data wrangling turns raw information into **tidy data** for structured insights

Sources

https://pandas.pydata.org/docs

https://aeturrell.github.io/python4DS

https://www.ibm.com/think/topics/data-wrangling

https://www.researchgate.net/publication/215990669_Tidy_data

Image Sources

[1] https://d33wubrfki0l68.cloudfront.net/6f1ddb544fc5c69a2478e444ab8112fb0eea23f8/91adc/images/tidy-1.png
[2] https://www.researchgate.net/publication/215990669_Tidy_data
[3] https://aeturrell.github.io/python4DS/data-tidy.html
[4] https://marketplace.visualstudio.com/items?itemName=ms-toolsai.datawrangler
[5] https://storage.googleapis.com/algodailyrandomassets/curriculum/standvsnorm/comparison_graph.png

Tutorial

https://github.com/justingebert/datawrangling-tidydata

Thank you